Jovana Todorović

Senior Software Engineer

Master of Information Systems graduate with 5.5 years of back-end engineering experience, specializing in distributed systems and microservices. Experienced in designing, developing, and operating production-grade services using Node.js, NestJS, Go, MongoDB, and PostgreSQL. Strong focus on scalability, reliability, observability, API design, and system ownership.

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- github.com/jtodorovic

SKILLS

JavaScript · TypeScript · Go · Python · Node.js · NestJS · Gin · Express · FastAPI · MongoDB · PostgreSQL · MySQL · Redis · DynamoDB · Mongoose · TypeORM · BullMQ · GraphQL · Docker · Kubernetes · AWS ECS · AWS ECR · AWS Lambda · AWS CloudWatch · S3 · MinIO · Azure Blob · Prometheus · Grafana · Loki · Sentry · JWT · Okta · Git

WORK EXPERIENCE

Senior Software Engineer

ThinkIT / Goin'

09/2023 - Present

- Developed and maintained a **Node.js** microservices backend (6 services) supporting approximately 200k active users with high reliability and scalability.
- Designed and implemented authentication and identity management features, including **JWT-based authentication**, social login providers (Google, Apple, Facebook, Microsoft Azure), **Okta** integration, email verification, phone verification, and SMS-based two-factor authentication (2FA).
- Designed event-driven workflows and background processing pipelines using **BullMQ** for notifications, user engagement, and email automation.
- Optimized **MongoDB** aggregations and scheduled workloads, significantly reducing execution time and database load.
- Implemented observability using **Prometheus, Grafana, Loki, and Sentry**, including distributed tracing across microservices, enabling proactive monitoring, performance analysis, and faster incident response.
- Built and maintained **CI/CD pipelines** using GitLab CI and **Docker**.
- Delivered **AI-powered** chat automation and analytics features and designed an achievement tracking service.

Back-end software engineer

Everseen d.o.o.

01/2021 - 09/2023

- Developed and maintained the back-end architecture of a data-intensive web application using **NestJS**, focusing on efficient data handling and performance optimization.
- Designed and implemented **RESTful** and **GraphQL** APIs as part of a microservices architecture.
- Implemented complex database queries, ETLs and migrations in **MongoDB** and **PostgreSQL**, handling over 1 million documents without data loss.
- Integrated third-party storage services such as **MinIO S3** and **Azure Blob**, for enhanced data management and scalability.
- Monitored and optimized API calls, resulting in significant latency improvement.
- Worked with containerized applications deployed on **Kubernetes**, assisting with deployment verification, troubleshooting, log analysis, and environment configuration.

EDUCATION

Master studies
Faculty of Organizational Sciences, University of Belgrade
10/2020 - 07/2022
GPA: 10.00

Bachelor studies
Faculty of Organizational Sciences, University of Belgrade
10/2016 - 09/2020
GPA: 9.86

PERSONAL PROJECTS

MacroTrackr **(2025 - Present)**
- Designed and built a back-end service for tracking macros intake using **Go (Gin framework)**.
- Containerized the application using **Docker** and deployed it to **AWS ECS (Fargate)**.
- Integrated **AWS** services: **ECR** for container registry, **CloudWatch** for centralized logging, **Lambda** for serverless routes, and **DynamoDB** for scalable data storage.
- Implemented a hybrid architecture combining REST endpoints with serverless functions.

MacroTrackr-AI **(2025)**
- Designed and implemented an AI-powered back-end service in **Python** using **FastAPI**.
- Integrated and evaluated multiple open-source **LLM models**, benchmarking performance and response quality to identify the optimal model for CPU-only execution environments.
- Implemented intelligent **Redis-based caching** for LLM responses to reduce repeated inference cost and improve response times.

Multi-layer Neural Network Implementation **(2025)**
- Implemented a custom **multi-layer neural network** from scratch; experimented with various hyperparameters including learning rate, number of layers, number of neurons per layer, and batch size.
- Tested different **activation functions** (ReLU, Sigmoid, Tanh) and analyzed their impact on convergence and performance.

Car Center **(2020 - 2022)**
- Independently analyzed real-world documentation and workflows, translating business processes into a structured software system and domain model; designed the overall system architecture and relational database schema.
- Implemented **Node.js (Express)** back-end and a hybrid front-end application using **Angular** and **Ionic**.

Geniefy - Travel Planner **(2020)**
- Participated in end-to-end system design, including application architecture, data modeling, and feature definition for a startup concept application.
- Designed and implemented a relational **SQL** database schema, ensuring data consistency and efficient querying.
- Implemented **Node.js (Express)** back-end and a hybrid front-end application using **Angular** and **Ionic**.

HONOR AWARDS

Best Students of the Generation (2017)
Faculty of Organizational Sciences

Best Students of the Generation (2018)
Faculty of Organizational Sciences

Best Students of the Generation (2019)
Faculty of Organizational Sciences

Best Students of the Generation (2020)
Faculty of Organizational Sciences

CERTIFICATES

Certificate of completion - „Data Science“ course (2019)

LANGUAGES

Serbian
Native or Bilingual Proficiency

English
Full Professional Proficiency